🛰️ SATELLITE EARTH OBSERVATION & LAND INTELLIGENCE

# SiteWise AI

Autonomous Land Cover Classification, Pattern-Affinitive Propagation & Multi-Criteria Site Suitability

**PAPNet Backbone**
**Pattern Affinity Network**

**EuroSAT Benchmark**
**10 Land Cover Classes**

**Sub-120MB Edge Cloud**
**Render 512MB RAM Ready**

Deep Learning Platform for Geospatial Zoning & Infrastructure Planning

# The Challenge: Land Assessment at Scale

Why manual GIS workflows and standard CNNs fail in modern satellite zoning

### ⏳ High Survey Latency

- Traditional field surveys take weeks to months
- Prohibitive cost per km² surveyed
- Static zoning maps lag real-time development
- Cannot keep pace with rapid urban sprawl

### 🧩 Disjoint Spatial Topologies

- Standard 3x3 CNN filters only see local pixels
- Earth observation features span non-local regions
- Fragmented crop parcels, river paths, and roads
- Requires long-range pattern affinity propagation

### 💾 Cloud Memory Barriers

- Full TensorFlow stack eats >600MB RAM on import
- Crashes cost-effective 512MB cloud free tiers
- Multi-worker servers multiply memory usage
- Demands an ultra-lightweight ONNX/LiteRT engine

**SiteWise AI solves all three: autonomous processing, non-local modeling, and sub-120MB execution.**

# End-to-End Multi-Tier AI Pipeline

Cascading multi-model inference from 10m raw satellite pixels to ranked municipal recommendations

### 1. Satellite Input

- Sentinel-2 MSI
- 64×64 RGB tiles
- 10m Ground Sample Dist.
- Multispectral Bands B2, B3, B4
- Auto-contrast normalization

### 2. Model 1 Backbone

- EuroSAT 10-Class model
- Dual Backbone Pool:
  - PAPNet (91.67%)
  - EfficientNetB0 (94.96%)
- 10 Softmax land probabilities
- Dual land/water detection

### 3. Visual Indices

- 8 Handcrafted indices:
  - Mean R, G, B
  - Brightness Index
  - Vegetation Index (NDVI)
  - Water Fraction (NDWI)
  - Built-up Index (NDBI)
  - Road Presence Marker

### 4. Decision Engines

- 18 Unified Input Features
- Model 2 (Facility Predictor):
  - Bank vs. School binary
- Model 3 (Site Recommender):
  - 8-profile suitability matrix
- Calibrated suitability rating

**Complete pipeline latency: < 50ms from raw pixel upload to multi-criteria recommendation.**

# PAPNet: Pattern-Affinitive Propagation

A tailored neural architecture capturing non-local spatial coherence across Earth observation tiles

## 📐 Mathematical Architecture

1. Projection to Latent Spaces:
   Q = Conv_q(X),  K = Conv_k(X),  V = Conv_v(X)

2. Pattern Affinity Matrix (Non-Local Correlation):
   A = Softmax( (Q · K^T) / √d )
   Computes pairwise semantic relatedness between all patches.

3. Feature Context Propagation:
   P = A · V

4. Learnable Residual Gating:
   Y = X + γ · Conv_proj(P)
   γ is initialized to 0.0 for stable convergence from step 1.

## 🛰️ Remote Sensing Advantages

• Disjoint Land Connectivity:
  Links separated agricultural plots and meandering river corridors across distant corners of the image.

• Multi-Scale Texture Harmony:
  Simultaneously handles high-frequency textures (residential buildings) and macro biomes (forest canopy).

• Lightweight Attention:
  Achieves global receptive field without the prohibitive quadratic complexity of vision transformers (ViTs).

• Fast Training:
  Reaches 91.67% accuracy in 15 epochs on EuroSAT.

**Custom Keras serializable layers ensure clean conversion to TFLite and ONNX formats.**

# Dual Land & Water Recognition

Eliminating shoreline and flood risk misclassifications through calibrated multi-spectral heuristics

## 🌊 Spectral Boundary & NDWI Modeling

• Standard classifiers force a single winner:
Treating hybrid riverbanks as either 100% forest or 100% water creates catastrophic planning errors.

• Calibrated Dual Detection:
- Checks total water probability (River + Lake/Sea ≥ 15%)
- Checks top water class confidence (≥ 12%)
- Checks water pixel fraction via NDWI (≥ 8%)

• Composite Classification:
Produces accurate composite labels: 'Forest with Water Body (River)' or 'Residential with Lake Basin'.

## 🛡️ Automated Hazard Mitigation

• Dynamic Flood Penalty:
Water presence automatically penalizes commercial and residential facility scores:
- Score -= 0.50 * River - 0.50 * Lake/Sea

• Buffer Zone Enforcement:
Protects riparian wetlands and waterways from illegal industrial runoff or toxic contamination.

• Realistic Multi-Criteria Ratings:
A plot with scenic river frontage gets high ratings for Green Conservation, but low ratings for heavy factories.

**Solves the notorious flat-water artifact problem and protects critical wetland ecosystems.**

# Empirical Results & Benchmark Performance

Rigorous evaluation across 27,000 Sentinel-2 multispectral tiles in the EuroSAT dataset

| 91.67% | 10 Classes | 8 Profiles | < 50ms |
| --- | --- | --- | --- |
| **PAPNet Accuracy (EuroSAT Test Set)** | **Land Cover Categories Crops, Forest, Urban, Road...** | **Site Recommender Multi-Criteria Head** | **Inference Latency Sub-second Web Response** |

Robust Confusion Matrix: High F1 scores across Annual Crops (0.93), Highway Corridors (0.99), and Industrial Zones (0.98) with zero false-positive bleed into water classes.

# Multi-Criteria Suitability Decision Matrix

Empirical validation showing how Model 2 and Model 3 score diverse satellite landscapes

| Satellite Terrain Tile | Model 2: Bank | Model 2: School | Top Recommended Use | Suitability Score |
|---|---|---|---|---|
| 🏢 Urban Residential | 100.0% (High) | 100.0% (High) | Residential Development | 100.0% (Optimal) |
| 🌲 Forest Canopy | 0.0% (Very Low) | 0.0% (Very Low) | Green / Conservation | 91.9% (Preserve) |
| 🏭 Industrial District | 100.0% (High) | 0.0% (Hazard Penalty) | Industrial Development | 100.0% (Optimal) |
| 🌾 Annual Crop Land | 0.0% (Very Low) | 0.0% (Very Low) | Agricultural Use | 99.0% (Agronomy) |
| 🛣️ Highway Corridor | 100.0% (High) | 100.0% (High) | Transportation Infra | 100.0% (Logistics) |

**Key takeaway: Schools receive 0% in industrial zones due to toxicity rules, while forests score 91.9% for conservation.**

# Cloud Optimization: Sub-120MB on Render

Overcoming the 512MB RAM barrier: 78% memory reduction using ONNX & LiteRT runtimes

## ❌ Standard TensorFlow Stack

- RAM Usage on Import: 606.6 MB
  (Instantly crashes Render 512MB free tier with OOM)
- 500MB+ TensorFlow wheel download
- 8–10 minute slow Docker build times
- Heavy CUDA/C++ shared library linking
- Multi-worker Gunicorn multiplies RAM usage

## ✅ Optimized ONNX / LiteRT Stack

- RAM Usage on Inference: 115.2 MB
  (Leaves >380MB of safe headroom on Render free tier)
- ~15MB lightweight runtime wheel
- < 30 second rapid build times
- Zero TensorFlow imports in production
- Gunicorn 1 worker + 4 threads shares model memory

**Result: 78% reduction in memory consumption enables free, reliable, production-grade 24/7 cloud deployment.**

# Interactive Ground Station HUD

React 18 single-page application with mission-control telemetry and radar reconnaissance animations

### 📡 Live Radar Telemetry

- Real-time laser scanning line beam
- 4-corner reticle sensor lock
- Live orbital telemetry stamps
- 3-stage visual inference pipeline
- Instant visual feedback

### ⚡ Instant Facility Recalculation

- Instant dropdown facility selector
- Immediate asynchronous re-scoring
- Dynamic engine badge indicator:
  - Model 2 (Bank / School binary)
  - Model 3 (8-profile matrix)
- Real-time confidence meters

### 📊 Spectral Telemetry Tabs

- 10-Class Softmax probability distribution
- 8 Visual indices telemetry bars:
  - Vegetation (NDVI)
  - Water fraction (NDWI)
  - Built-up index (NDBI)
- Full ranked recommendation list

**Built with React 18, Tailwind CSS, Lucide icons, and responsive glassmorphism HUD design.**

# Future Vision & Next Steps

Scaling SiteWise AI toward municipal cadastral integration and multi-sensor Earth observation

### 🛰️ Sentinel-1 SAR Fusion

- Synthetic Aperture Radar (SAR)
- All-weather day/night penetration
- Eliminates cloud cover occlusion
- Dual optical-SAR feature fusion
- Enhanced flood & moisture maps

### 📈 Temporal Change Tracking

- Time-series satellite comparison
- Automated deforestation detection
- Urban sprawl & construction tracking
- Seasonal crop yield predictions
- Disaster recovery monitoring

### 🏛️ Cadastral & Municipal APIs

- Direct integration with city registries
- Automated building code compliance
- Environmental ESG reporting
- Real-time parcel suitability scoring
- Public planning portal embeds

**SiteWise AI: Precision Earth Observation Intelligence for Sustainable Development**

GitHub: https://github.com/GarvKapoor/site-wise-ai-papnet • Live: https://site-wise-ai-papnet.onrender.com